Devising Scrutable User Models for Time Management Assistants

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ABSTRACT

Intelligent Personal Assistants (IPAs) have become ubiquitous through integration into smartphones, smart speakers, and standalone devices. However, prior studies raised noteworthy usability concerns and determined that IPAs remain primarily used for simple tasks. Such findings contrast with the reported user aspirations for more proactive and truly personalised IPAs. By focusing on the use case of time management, in this paper, we contemplate how scrutability – i.e. the ability of the user to study their assistant and its underlying user model – fits within the vision of more complex IPAs. Furthermore, we describe our ongoing project investigating user interest in and expectations of the scrutability of a proactive calendaring assistant. Lastly, by deliberating on the challenges and benefits of making IPAs scrutable, this paper outlines potential avenues for further research.

CCS CONCEPTS

• **Human-centered computing → Personal digital assistants**; *User models*; *User centered design.*

KEYWORDS

Intelligent personal assistants, Explainability, Time management, Proactive dialogue systems

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1 INTRODUCTION

Intelligent Personal Assistants (IPAs), also referred to as digital agents or virtual assistants, have become a standard feature of contemporary smartphones and computers, as well as the main interaction interfaces of devices such as smart speakers and smart home controllers [6, 15, 38, 43]. The application areas in which IPAs have been applied range from education [8, 30] over lifestyle and cognitive assistance [11, 40] to health-related applications [14, 39]. The IPAs with which the user interacts primarily – or exclusively – using voice are also known as voice assistants and are a type of Conversational User Interface (CUI) [38]. Some systems also offer multimodal communication with the user [30]. The most prominent commercially offered assistants, such as Siri and Google Assistant, are voice assistants that also enable (and sometimes necessitate) the user to interact with them through a Graphical User Interface (GUI) [12, 42].

This proliferation of IPAs inspired user studies investigating human-assistant interaction, which revealed usability limitations [2, 15]. IPAs were found to be used primarily for straightforward, functional tasks such as performing one-step actions (e.g. start music, switch a device on or off), finding information (e.g. relay a weather forecast), or providing simple social interactions (e.g. tell a joke or a short story) [31, 46]. Prior studies have also indicated that current IPAs might deteriorate the user's multi-tasking abilities [1, 19], rather than support them in their knowledge work as had been envisioned in pioneering IPA works [5, 35]. Further causes of user dissatisfaction with their IPAs include misunderstanding their requests or intent, inappropriate modality choice (using voice in public or asking the user to look at the phone while driving), and a lack of transparency regarding the system's capabilities [12, 31]. Concerns regarding privacy and security have also been reported in several user studies [6, 12, 45], with Alizadeh et al. [2] noting that lack of explanations for IPA behaviour further frustrates the users. The current generation of IPAs is also primarily reactive, typically waiting for the user to activate them and then responding to an explicit prompt [12, 32]. The proactive capabilities of Siri and Google Assistant are limited to providing pieces of information on the screen deemed relevant for the user [22, 42]. In contrast, user studies [32, 46, 47] indicate that the users want future IPAs to be more proactive, anticipating their needs and acting accordingly. This proactivity would also require a higher level of personalisation than what is currently the norm, necessitating that the IPA builds and maintains a user model [22, 32].

Scrutability is the ability of the user to actively study the underlying user models towards gaining an understanding of how the personalisation is achieved [26]. Scrutability has been identified as a key characteristic of the UMAP community's research [16], and it was extensively investigated in domains such as education [26, 28], ontology overview [26], and recommender systems [3, 36]. It was proposed that enabling scrutiny might mitigate previously identified human-assistant interaction challenges and allow for a transition from predominantly reactive to more proactive and personalised assistance, but research on scrutable human-IPA interaction remains scarce [22, 23]. In this ongoing project description, we offer an overview of the prior work, recognised challenges, and foreseeable opportunities in defining the principles of scrutability for IPAs. In particular, we look at time management assistance, a complex yet bounded use case for the envisioned proactive IPAs [23, 46, 47]. We conclude by outlining our planned future work and

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indicating potentially fruitful avenues for further Explainable User Modelling (ExUM) exploration.

2 RELATED WORK

2.1 Anticipated Human-Assistant Interaction

In their IPA literature review, de Barcelos Silva et al. [15] identified personalisation and proactivity as a future research challenge. Similarly, a study by Völkel et al. [46] found that participants envisioned that perfect future IPAs would have knowledge about them and their environment and would use this knowledge to offer proactive suggestions and recommendations. The study also reported different opinions on the desired degree of humanness in the dialogue and the conversational role of the IPA (ranging from purely functional to the one of a friend). In a qualitative analysis, Doyle et al. [18] found that user perceptions of humanness are multidimensional and affected by vocal qualities and perceptions of IPA capabilities. Zargham et al. [47] investigated the perceived appropriateness of proactive IPA interventions, focusing on voice assistants akin to present-day smart speakers. They found that the users had varying levels of desire for proactive behaviour overall and a strong correlation between the appropriateness, usefulness, and incisiveness of proactive interventions. The three most useful proactive interventions were to have the IPA 1) call the fire department after detecting a fire; 2) alert the user that they must not snooze the alarm again, so as not to miss a meeting; and 3) offer to arrange a doctor's appointment when the user coughs. Stepping up from voice-only interactions, Këpuska and Bohouta [30] proposed a structure of next-generation IPAs for multimodal input and output. The structure includes gesture understanding, real-time video analysis, as well as responding to the user both vocally and with visualisations.

Personalised time management assistance was one of the main use cases for CALO, an IPA development project, whose spin-out was Siri [5]. The system was designed to help with the reservation of resources, scheduling meetings, and even communicating the user's needs in semi-automated negotiations [5]. The CALO user would enter their guidance and control system restrictions through declarative commands [35]. More recently, Cranshaw et al. [13] introduced Calendar.HELP, a micro-services framework built for hybrid intelligence that delegates meeting scheduling negotiations via email. In an analysis of expert opinions regarding time management IPAs, Jeromela and Conlan [23] detail envisioned challenges and use cases. The use cases included assisting the user to reflect on and improve their time planning, nudging the user towards healthy behaviour, and partaking in multiparty negotiations and information sharing. The experts in the study [23] voiced concerns about the user's ability to appropriately estimate the system's capability and their willingness to share sensitive data, thereby echoing the warnings from prior studies [6, 12].

2.2 Principles of Scrutability

Scrutability, a term initially used by Kay [25], denotes the ability of the user to actively probe and study user models of a personalised system. It should be noted that Tintarev and Masthoff [44] give a diverging definition of scrutability and equate it with the ability of the user to modify the modelled information. In line with Kay and Kummerfeld [26] and subsequent works on scrutability [16, 22, 23, 28], this paper sees modifying user models as an element of *user control*, that is enabled and preceded by, but by definition excluded from, scrutability. As Khosravi et al. [28] elaborate, explainability is another term closely related to scrutability. Yet, the former typically focuses on the ability of the system to generate justifications or interpretations for its behaviour. In contrast, the latter emphasises that the user may delve into the working of the system towards gaining a (partial) understanding of aspects of the system [23, 28].

Kay and Kummerfeld [26] propose four principles for building scrutable user models: 1) *parallel design* (the interface and the user model representation and reasoning are considered together throughout the design process); 2) *first-class citizen* (a centralised user model for all personalised applications and a unified control and scrutiny interface are provided); 3) a *context-based interpretation* is supported; and 4) *client-side* (the user model is stored in a manner which guarantees full user access and control over the stored data). Thus, the principles of first-class citizens and client-side explicitly call for a paradigm shift of the current standard practice, where user models are mostly application-specific and stored on servers inaccessibly for the user [26]. Scrutability and the four principles address the problems regarding *privacy*, *invisibility* of the influence of personalisation in displayed content, *errors in user model*, *wasted user models* (not utilising data for personalisation), and *the lack of control*. In a preliminary interaction framework for scrutable IPAs, Jeromela [22] has raised the need for extending the principles of scrutability due to novel problems that proactive and delegative IPAs present. A similar undertaking in the area of Personalised User models for life-long Learners (PUMLs) has resulted in an application-specific set of competency questions and design guidelines [27].

3 DOMAIN-SPECIFIC CHALLENGES AND SCRUTABILITY

IPAs that always listen and can proactively engage based on context and user intent [23, 43, 46] raise a significantly different set of considerations relative to systems – such as hypertext editors and recommenders [26] – for which scrutability was previously considered [22]. This indicates that the defining problems and designing principles of scrutability, as presented in [26] and understood by the UMAP community [16], might need to be revisited and extended. This section describes some of the challenges of time management IPAs and how they relate to the role of scrutability.

3.1 Interaction Design

Proactive IPAs for time management, as envisioned by the experts in [23], are complex proactive systems capable of initiating and leading conversations with the user multimodally (via GUIs and CUIs). It was also envisioned that they might ultimately substitute the user in tasks such as meetings or task allocation negotiations. Among the implications of such complex use cases are the six interaction-related challenges detailed below.

Proactivity. Based on the use cases the experts in [23] and the participants in [46] envisioned, proactive behaviour of time management assistants can be triggered by 1) the current time or scheduled events (alarms and appointment reminders); 2) the activities the user is engaged in (be it if they are busy, so the IPA chooses to postpone an engagement or because they are free so an opportune time for raising a non-urgent matter is identified); 3) the environment the user is in (whether an engagement is appropriate for the surroundings of the user); 4) and other people's needs (e.g. an attendance confirmation request). The experts in [23] suggested that the IPA may proactively nudge the user to scrutinise some of its decisions and models at opportune moments. Several prior studies, such as [29, 47], had already looked into the acceptability of proactive agent behaviour, but further research is needed to establish how scrutability would affect the user attitudes towards proactive IPAs and the receptiveness towards "scrutiny nudges".

Delegation. Having the IPA substitute the user in certain actions is another envisioned use case [22, 23, 46]. For instance, an expert in [23] proposed the IPA might delegate scheduling negotiations and event information dissemination, a use case similar to Calendar.Help [13]. More challengingly, the experts also proposed for the IPA to take over task allocation communications (specifically, on agreeing whether they or their spouse would be the one to take their child to an extracurricular activity). Delegation presents a challenge for the facilitation of scrutability because, in providing the information, the IPA has to communicate how the decision was made and why it was judged to be most appropriate or fair while not revealing information that the other parties in the negotiation would not want to have shared. This suggests further work might continue from approaches that personalise explanations, cf. Chambers et al. [10], and adapt them for scrutiny of multi-user collaborative scenarios in a privacy-preserving manner.

Modality Selection. Previously investigated scrutable applications relied primarily on a GUI [22]. Contrastingly, experts interviewed by Jeromela and Conlan [24] envisioned time management assistants as multimodal systems that would be able to interact with the user using both a GUI and a CUI and would be able to change the response mode proactively. This poses a two-fold challenge for the realisation of scrutability. On the one hand, it should be determined how the IPA is to facilitate the scrutiny if the required information is more straightforward to display visually (e.g. how the assistant sees the user's availability over the upcoming period), but the prior communication was vocal. On the other hand, as experts in [23] further deliberate, the user must be able to scrutinise and control what classes of information the IPA might proactively share vocally and under which circumstances. Otherwise, the user might be hesitant to share any sensitive or potentially embarrassing information with the assistant due to privacy concerns.

Humanness and Expediency. Prior studies, i.a. [12, 18, 31], have looked into the perception and desirability of the humanness of IPAs (including the perceived personality, humour, and empathy). However, the IPAs considered in these studies typically have non-adaptive responses and are incapable of providing meaningful explanations as to why a particular trait was displayed. Allowing the user to scrutinise and control the "humanness" of the assistant may thus improve interaction satisfaction. Prospective (ante-hoc) methods, championed by Shneiderman [41, Ch. 19], could be integrated here as well, allowing the user to set the desired level of humour and empathy of the IPA. More tightly related to the use case of time management is the expediency of the assistant's feedback. Dennis et al. [17] reported that adapting feedback to the personality of learners might benefit their performance. Likewise, allowing the user to set their desired feedback (say, on a spectrum ranging from just giving the performance statistics to providing human-like praise) might support the envisioned [24] goal of facilitating and supporting reflection on time use.

Disclosure of Capabilities. As was described by Jeromela and Conlan [24], the interviewed experts underlined the necessity for the assistant to disclose the extent of its capabilities accurately. Mishra and Rzeszotarski [33] describe factors influencing user expectations of interactive interfaces, highlighting the role of user experience with the task at hand and program type, as well as the aesthetics of the interface. Similarly, Moore [34] argued that the human-like aspects or the dialogue capabilities of the voice assistant should match the extent of its actual functionalities to give the user a fitting intuition. Contrastingly, state-of-the-art IPAs as well as chatbots based on large language models, such as ChatGPT, may lead the participants to overestimate their capabilities, as humanness sets unrealistic expectations [4, 12, 33]. Prior studies have indicated that an accurate view of the system's capabilities (or the user's model of the assistant) was seen as tightly related to usage willingness and the user's openness to "forgive" the IPA's mistakes [12, 23, 31]. Thus, further studies might look into how scrutability can aid the user in understanding the IPA's capabilities.

Effort-Efficiency. In prior studies on scrutable applications (e.g. on text editors [26] or recommender systems [3, 36]), the only way to perform the task was to use the provided application. In contrast, with time management assistants or IPAs in general, the user can always do the task "manually", without any assistance. Given that scrutability is a prerequisite to user control but requires that the user invests time in understanding the system first, it remains to be investigated how scrutability affects usage willingness. Thus, as Jeromela [22] in reference to Völkel et al. [46] previously proposed, it might be worthwhile investigating how scrutable IPAs can be effort-efficient, that is, how to achieve that their use is principally less time-consuming for the user than just doing all tasks without them.

3.2 User, Context, and Risk Modelling

Scrutable user models allow the user to investigate and control the manner in which the data is captured, stored, and used to facilitate personalised services [26]. In this paper, we do not look into specific technologies or approaches to user modelling but explore the challenges that need to be addressed for the IPA and its models to be considered scrutable.

Meaningful Control. The reach of the envisioned IPA capabilities [23, 46], such as proactively modifying events in the user's calendar, partaking in negotiations, or spending money to book tickets, reinforces the need for user control of IPAs. As previously proposed proactive IPA frameworks [22, 23, 32] also envisioned that the IPA would model not only the user, but also the surrounding context and the potential risk of its actions, further research should investigate how scrutiny and control of these models can be facilitated. A metric to consider in this pursuit is user trust, which was previously shown to be affected by factors including predictability and understandability [21, 29, 37, 45].

Minimal Interpretability. The ability of the user to scrutinise the IPA's actions is a prerequisite to enabling user control [26]. However, as argued in [23], some forms of machine learning algorithms are inherently opaque. How, or even if, such machine learning systems can be made scrutable is a manner of active research. In that pursuit, we see value in both the algorithm-first ("opening the black box" explanations) as well as the human-centric approach, which aims to understand user perception and expectations. To this end, a meaningful aim may be to define a "minimal" set of requirements for interpretability that an IPA must facilitate for it and its models to be considered scrutable.

Data Stewardship. Both users [12, 43] and experts [6] have expressed concerns regarding the manner in which IPAs collect, store, and share their data. Tabassum et al. [43] found users to be more willing to share a conversation with an assistant if the content of the conversation was less sensitive and the service provided in return more tangible. As reported by Jeromela and Conlan [23], time management data is inherently sensitive because it deals directly with one's daily schedule and social contacts (a sentiment directly echoed by a participant in [43]). To attempt to remedy this concern, the idea of *Stages* is introduced in [23]. Each *Stage* refers to a set of IPA functionalities that is explicitly connected to a set of specific data licences, which allow the assistant to collect and use a class of data to facilitate these functionalities. We hypothesise that facilitating *Stages* might comprise a novel scrutability principle, mitigating data stewardship concerns while facilitating user control and revealing the extent of IPA capabilities [23].

4 PROPOSED METHOD

In this section, we outline details of our two upcoming user studies that investigate how the principles of scrutability can be extended to facilitate the challenges described in the previous section.

4.1 Task-Based Study on Scrutability of a Proactive Calendaring Assistant

Our next study, currently undergoing an ethical review, has three segments. The participants will first fill in a questionnaire, followed by a task-based interaction with a simple prototype of a scrutable time management assistant. The study will conclude with a focus group discussion on interaction impressions and views on the proactivity and scrutability of IPAs. The questionnaire will have the participants indicate their age, level of education, prior experiences with and opinions of IPAs, and a typical weekly schedule to be taken as the basis for their user model. The tasks will include calendar manipulation, time use reflection, and model scrutiny. We will use a proprietary GUI as a proxy for the IPA.

The prototype has a rather straightforward design, with the majority of the screen dedicated to the workspace that offers a visualisation most appropriate for the context. The default context is a calendar view of the user's monthly schedule. When creating a new event, this changes to a weekly overview. To facilitate scrutiny, the workplace also offers visualisations that follow the PUMLs design guidelines to accompany textual justifications [27]. For instance, should the user want to know why a certain date was recommended for an activity, the assistant might state that this was judged to be the least busy day and offer a colour-coded overview of the week. Then, the user may enquire further, for instance, by looking into how the IPA defines business (which calendar events are weighted in) or by inquiring into *Data Stewardship* more generally. The user could also *Control* the IPA's models by changing how events are classified. Similarly as in [45], the primary mode of interaction for the participant is through on-screen buttons, one of which allows them to start a multi-layered iterative process of scrutiny. This relatively simple design simplifies the challenge of *Disclosure of Capabilities*, as users always know what the possible next steps are.

While the study design does not allow for an investigation into *Delegation* and *Modality Selection*, the tasks and focus group prompts will delve into *Proactivity*, and *Data Stewardship* through *Stages*. Namely, the prototype enables the user to choose between three *Stages*: Clash-Checking Only, where the proactivity is most limited; Default, where the IPA also dynamically suggests event placements or edits depending on the relevant data in the user model; and the Active Restructuring, where the IPA prototype edits or moves events and then notifies the participant of the change after-the-fact. In each such case, the proactive actions of the IPA are indicated to the user through text displayed above the menu. The proactive actions and text follow programmed rules that look into calendar events and the information about user preferences stemming from the questionnaire and prototype interaction. Some of the proactively displayed text will also invite the participant to scrutinise the personalised actions the assistant offers.

As a recent study on explanation preferences [9] has found that there might be a disconnect between the participants' claims and their measured performance, our analysis is to include both qualitative and quantitative methods. Specifically, we intend to perform a thematic analysis (a qualitative research approach described by Braun and Clarke [7] and often used in human-assistant interaction studies [12, 23, 31]) of the transcripts recorded during the focus group discussions and the individual prototype interactions (the think-aloud approach [20]). To facilitate quantitative analysis, every option the user clicks, as well as the IPA's advice text and system responses, will be logged. These logs will then be analysed to investigate whether the users had shown interest in scrutinising the assistants (if they unpromptedly visited the "about" or "scrutinise" and how deeply they engaged in the iterative process of studying the underlying models and information contained). Possible correlations with the data from the questionnaire will also be investigated.

4.2 Longitudinal Study with a Scrutable Time Management Assistant for Students

The subsequent user study will build on findings from the task-based study from the previous subsection. So far, we envision this time management IPA prototype as a browser extension to be offered to students of the authors' university over an academic semester (allowing for a longitudinal, in-the-wild perspective). The extension is planned to connect with the API of the university's eLearning platform (through which lecture dates and deadlines are shared) and Google Calendar. The browser will then allow the participating students to enter or modify events in their calendars from the eLearning platform. The assistant is envisioned to offer scrutable, proactive, and personalised advice. For instance, it might remind the student to start working on the assignment if there is a gap in their schedule on a particular day and the student had previously indicated that they want such a nudge to happen. The type of advice and the form of delivery could be changed, in accordance with the challenge of *Humanness and Expediency*. As the prototype would offer time management services regarding real-life tasks, the rate of its use will offer us an insight into *Effort-Efficiency*. The study will end with participant interviews to be analysed alongside collected anonymised usage logs.

5 LIMITATIONS AND FUTURE WORK

The proposed studies also have several limitations. Most notably, both prototypes are purely textual and computer-based. In contrast, the current [15] and the envisioned [46] IPAs are typically multimodal and primarily mobile, barring an investigation into challenges of *Modality Selection* and scrutability of more complex context and risk modelling. The task-based study also has a very constrained dialogue design, preventing us from investigating the *Humanness* (apart from providing personalised feedback and suggestions). In fact, because the prototypes to be used in the studies are more akin to an eCalendar and a browser extension than to commercially available assistants such as Siri, the participants in our studies might disassociate them from IPAs they are already familiar with. This is a significant limitation, as prior studies have shown that familiarity with a type of intervention might affect the user acceptance of proactive actions [47] and the level of trust [37]. Lastly, both prototypes are planned to be based on relatively straightforward rule-based systems, and further research will be needed to investigate how the insights on scrutability translate to systems reliant on opaque algorithms (i.e. defining the *Minimal Interpretability* requirements).

Alternative avenues of research arising from identified challenges include the scrutability of IPAs with richer communication abilities, be it through the use of more advanced chatbots or the Wizard-of-Oz method. Delegation of communication, both in terms of technical realisation and user acceptability, remains scarcely researched, regardless of prior studies indicating user interest [23, 46]. Regarding the use case of time management more closely, more targeted research might investigate the role of scrutable IPA support for behavioural change. We believe that the challenges described in Section 3 offer useful railings for such research. However, we acknowledge that there might be further areas of interest at the confluence of scrutability, proactive IPAs, and time management. We thus also welcome adjustments or extensions of our proposed encapsulation of interaction and modelling challenges.

6 CONCLUSION

In this ongoing project paper, we considered the benefits and challenges in devising scrutable IPAs for time management. In doing so, we outlined plans for two upcoming user studies and proposed alternative avenues of research. These considerations may inspire and guide members of the explainable user modelling community to tackle the broader challenge of making IPAs more personalised and scrutable.

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REFERENCES

[1] Shashank Ahire, Aaron Priegnitz, Oguz Önbas, Michael Rohs, and Wolfgang Nejdl. 2021. How Compatible is Alexa with Dual Tasking? — Towards Intelligent Personal Assistants for Dual-Task Situations. In *Proceedings of the 9th International Conference on Human-Agent Interaction* (Virtual Event, Japan) *(HAI '21)*. Association for Computing Machinery, New York, NY, USA, 103–111. https://doi.org/10.1145/3472307.3484165

[2] Fatemeh Alizadeh, Dominik Pins, and Gunnar Stevens. 2023. User-friendly Explanatory Dialogues. https://doi.org/10.18420/muc2023-mci-ws16-120,https://dl.gi.de/handle/20.500.12116/42137

[3] Öznur Alkan, Massimiliano Mattetti, Elizabeth M. Daly, Adi Botea, Inge Vejsbjerg, and Bart Knijnenburg. 2021. IRF: A Framework for Enabling Users to Interact with Recommenders through Dialogue. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 163 (apr 2021), 25 pages. https://doi.org/10.1145/3449237

[4] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (Virtual Event, Canada) *(FAccT '21)*. Association for Computing Machinery, New York, NY, USA, 610–623. https://doi.org/10.1145/3442188.3445922

[5] Pauline M. Berry, Thierry Donneau-Golencer, Khang Duong, Melinda Gervasio, Bart Peintner, and Neil Yorke-Smith. 2017. Evaluating intelligent knowledge systems: experiences with a user-adaptive assistant agent. *Knowledge and Information Systems* 52, 2 (2017).

[6] Tom Bolton, Tooska Dargahi, Sana Belguith, Mabrook S. Al-Rakhami, and Ali Hassan Sodhro. 2021. On the Security and Privacy Challenges of Virtual Assistants. *Sensors* 21, 7 (2021), 2312. https://www.mdpi.com/1424-8220/21/7/2312

[7] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101.

[8] Laura Butler and Louise Starkey. 2024. OK Google, help me learn: an exploratory study of voice-activated artificial intelligence in the classroom. *Technology, Pedagogy and Education* 33, 2 (2024), 135–148. https://doi.org/10.1080/1475939X.2024.2311779 doi: 10.1080/1475939X.2024.2311779.

[9] Lenart Celar and Ruth M. J. Byrne. 2023. How people reason with counterfactual and causal explanations for Artificial Intelligence decisions in familiar and unfamiliar domains. *Memory & Cognition* 51, 7 (2023), 1481–1496. https://doi.org/10.3758/s13421-023-01407-5

[10] Owen Chambers, Robin Cohen, Maura R. Grossman, and Queenie Chen. 2022. Creating a User Model to Support User-specific Explanations of AI Systems. In *Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization* (Barcelona, Spain) *(UMAP '22 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 163–166. https://doi.org/10.1145/3511047.3537678

[11] Angelo Costa, Paulo Novais, and Vicente Julian. 2018. *A Survey of Cognitive Assistants.* Springer International Publishing, 3–16.

[12] Benjamin R. Cowan, Nadia Pantidi, David Coyle, Kellie Morrissey, Peter Clarke, Sara Al-Shehri, David Earley, and Natasha Bandeira. 2017. "What Can i Help You with?": Infrequent Users' Experiences of Intelligent Personal Assistants. In *Proceedings of the 19th International Conference on Human-Computer Interaction with Mobile Devices and Services* (Vienna, Austria) *(MobileHCI '17)*. Association for Computing Machinery, New York, NY, USA, Article 43, 12 pages.

[13] Justin Cranshaw, Emad Elwany, Todd Newman, Rafal Kocielnik, Bowen Yu, Sandeep Soni, Jaime Teevan, and Andrés Monroy-Hernández. 2017. Calendar.Help: Designing a Workflow-Based Scheduling Agent with Humans in the Loop. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) *(CHI '17)*. Association for Computing Machinery, New York, NY, USA, 2382–2393.

[14] Rachel G. Curtis, Bethany Bartel, Ty Ferguson, Henry T. Blake, Celine Northcott, Rosa Virgara, and Carol A. Maher. 2021. Improving User Experience of Virtual Health Assistants: Scoping Review. *J Med Internet Res* 23, 12 (2021), e31737. https://doi.org/10.2196/31737

[15] Allan de Barcelos Silva, Marcio Miguel Gomes, Cristiano André da Costa, Rodrigo da Rosa Righi, Jorge Luis Victoria Barbosa, Gustavo Pessin, Geert De Doncker, and Gustavo Federizzi. 2020. Intelligent personal assistants: A systematic literature review. *Expert Systems with Applications* 147 (2020), 113193.

[16] Paul De Bra. 2017. After Twenty-Five Years of User Modeling and Adaptation...What Makes us UMAP?. In *Proceedings of the 25th Conference on User*

Modeling, Adaptation and Personalization (Bratislava, Slovakia) *(UMAP '17)*. Association for Computing Machinery, New York, NY, USA, 1. https://doi.org/10.1145/3079628.3079662

[17] Matt Dennis, Judith Masthoff, and Chris Mellish. 2012. Adapting Performance Feedback to a Learner's Conscientiousness. In *User Modeling, Adaptation, and Personalization*, Judith Masthoff, Bamshad Mobasher, Michel C. Desmarais, and Roger Nkambou (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 297–302.

[18] Philip R. Doyle, Justin Edwards, Odile Dumbleton, Leigh Clark, and Benjamin R. Cowan. 2019. Mapping Perceptions of Humanness in Intelligent Personal Assistant Interaction. In *Proceedings of the 21st International Conference on Human-Computer Interaction with Mobile Devices and Services* (Taipei, Taiwan) *(MobileHCI '19)*. Association for Computing Machinery, New York, NY, USA, Article 5, 12 pages. https://doi.org/10.1145/3338286.3340116

[19] Justin Edwards, He Liu, Tianyu Zhou, Sandy J. J. Gould, Leigh Clark, Philip Doyle, and Benjamin R. Cowan. 2019. Multitasking with Alexa: How Using Intelligent Personal Assistants Impacts Language-Based Primary Task Performance. In *Proceedings of the 1st International Conference on Conversational User Interfaces* (Dublin, Ireland) *(CUI '19)*. Association for Computing Machinery, New York, NY, USA, Article 4, 7 pages. https://doi.org/10.1145/3342775.3342785

[20] Mark C Fox, K Anders Ericsson, and Ryan Best. 2011. Do procedures for verbal reporting of thinking have to be reactive? A meta-analysis and recommendations for best reporting methods. *Psychological bulletin* 137, 2 (2011), 316.

[21] Alyssa Glass, Deborah L. McGuinness, and Michael Wolverton. 2008. Toward Establishing Trust in Adaptive Agents. In *Proceedings of the 13th International Conference on Intelligent User Interfaces* (Gran Canaria, Spain) *(IUI '08)*. Association for Computing Machinery, New York, NY, USA, 227–236. https://doi.org/10.1145/1378773.1378804

[22] Jovan Jeromela. 2022. Scrutability of Intelligent Personal Assistants. In *Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization* (Barcelona, Spain) *(UMAP '22)*. Association for Computing Machinery, New York, NY, USA, 335–340.

[23] Jovan Jeromela and Owen Conlan. 2023. Onboarding Stages and Scrutable Interaction: How Experts Envisioned Explainability in Proactive Time Management Assistants. In *Proceedings of the 11th International Conference on Human-Agent Interaction* (Gothenburg, Sweden) *(HAI '23)*. Association for Computing Machinery, New York, NY, USA, 306–315. https://doi.org/10.1145/3623809.3623851

[24] Jovan Jeromela and Owen Conlan. 2023. Voicing Suggestions and Enabling Reflection: Results of an Expert Discussion on Proactive Assistants for Time Management. In *Proceedings of the 5th International Conference on Conversational User Interfaces* (Eindhoven, Netherlands) *(CUI '23)*. Association for Computing Machinery, New York, NY, USA, Article 48, 6 pages. https://doi.org/10.1145/3571884.3604317

[25] Judy Kay. 1999. *A scrutable user modelling shell for user-adapted interaction.* Doctoral dissertation. University of Sydney.

[26] Judy Kay and Bob Kummerfeld. 2013. Creating Personalized Systems That People Can Scrutinize and Control: Drivers, Principles and Experience. *ACM Trans. Interact. Intell. Syst.* 2, 4, Article 24 (jan 2013), 42 pages.

[27] Judy Kay and Bob Kummerfeld. 2019. From data to personal user models for life-long, life-wide learners. *British Journal of Educational Technology* 50, 6 (2019), 2871–2884. https://doi.org/10.1111/bjet.12878 arXiv:https://bera-journals.onlinelibrary.wiley.com/doi/pdf/10.1111/bjet.12878

[28] Hassan Khosravi, Simon Buckingham Shum, Guanliang Chen, Cristina Conati, Yi-Shan Tsai, Judy Kay, Simon Knight, Roberto Martinez-Maldonado, Shazia Sadiq, and Dragan Gašević. 2022. Explainable Artificial Intelligence in education. *Computers and Education: Artificial Intelligence* 3 (2022), 100074.

[29] Matthias Kraus, Nicolas Wagner, and Wolfgang Minker. 2020. Effects of Proactive Dialogue Strategies on Human-Computer Trust. In *Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization* (Genoa, Italy) *(UMAP '20)*. Association for Computing Machinery, New York, NY, USA, 107–116. https://doi.org/10.1145/3340631.3394840

[30] Veton Këpuska and Gamal Bohouta. 2018. Next-generation of virtual personal assistants (Microsoft Cortana, Apple Siri, Amazon Alexa and Google Home). In *2018 IEEE 8th Annual Computing and Communication Workshop and Conference (CCWC)*. 99–103. https://doi.org/10.1109/CCWC.2018.8301638

[31] Ewa Luger and Abigail Sellen. 2016. "Like Having a Really Bad PA": The Gulf between User Expectation and Experience of Conversational Agents. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) *(CHI '16)*. Association for Computing Machinery, New York, NY, USA, 5286–5297. https://doi.org/10.1145/2858036.2858288

[32] Christian Meurisch, Maria-Dorina Ionescu, Benedikt Schmidt, and Max Mühlhäuser. 2017. Reference model of next-generation digital personal assistant: integrating proactive behavior. In *Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers*. 149–152.

[33] Swati Mishra and Jeffrey M Rzeszotarski. 2023. Human Expectations and Perceptions of Learning in Machine Teaching. In *Proceedings of the 31st ACM Conference on User Modeling, Adaptation and Personalization* (Limassol, Cyprus) *(UMAP '23)*. Association for Computing Machinery, New York, NY, USA, 13–24. https://doi.org/10.1145/3565472.3595612

[34] Roger K. Moore. 2017. *Is Spoken Language All-or-Nothing? Implications for Future Speech-Based Human-Machine Interaction.* Springer Singapore, Singapore, 281–291. https://doi.org/10.1007/978-981-10-2585-3_22

[35] D. Morley and K. Myers. 2004. The SPARK agent framework. In *Proceedings of the Third International Joint Conference on Autonomous Agents and Multiagent Systems, 2004. AAMAS 2004.* 714–721.

[36] Cataldo Musto, Fedelucio Narducci, Marco Polignano, Marco de Gemmis, Pasquale Lops, and Giovanni Semeraro. 2020. Towards Queryable User Profiles: Introducing Conversational Agents in a Platform for Holistic User Modeling. In *Adjunct Publication of the 28th ACM Conference on User Modeling, Adaptation and Personalization* (Genoa, Italy) *(UMAP '20 Adjunct)*. Association for Computing Machinery, New York, NY, USA, 213–218. https://doi.org/10.1145/3386392.3399298

[37] Danielle Marie Olson and Yan Xu. 2021. Building Trust Over Time in Human-Agent Relationships. In *Proceedings of the 9th International Conference on Human-Agent Interaction* (Virtual Event, Japan) *(HAI '21)*. Association for Computing Machinery, New York, NY, USA, 193–201. https://doi.org/10.1145/3472307.3484178

[38] Martin Porcheron, Joel E. Fischer, Stuart Reeves, and Sarah Sharples. 2018. Voice Interfaces in Everyday Life. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–12.

[39] Samira Rasouli, Moojan Ghafurian, and Kerstin Dautenhahn. 2022. Students' Views on Intelligent Agents as Assistive Tools for Dealing with Stress and Anxiety in Social Situations. In *Proceedings of the 10th International Conference on Human-Agent Interaction* (Christchurch, New Zealand) *(HAI '22)*. Association for Computing Machinery, New York, NY, USA, 23–31. https://doi.org/10.1145/3527188.3561932

[40] Emre Sezgin, Lisa K Militello, Yungui Huang, and Simon Lin. 2020. A scoping review of patient-facing, behavioral health interventions with voice assistant technology targeting self-management and healthy lifestyle behaviors. *Translational Behavioral Medicine* 10, 3 (2020), 606–628.

[41] Ben Shneiderman. 2022. *Human-Centered AI.* Oxford University Press. https://doi.org/10.1093/oso/9780192845290.001.0001

[42] Yu Sun, Nicholas Jing Yuan, Yingzi Wang, Xing Xie, Kieran McDonald, and Rui Zhang. 2016. Contextual Intent Tracking for Personal Assistants. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (San Francisco, California, USA) *(KDD '16)*. Association for Computing Machinery, New York, NY, USA, 273–282. https://doi.org/10.1145/2939672.2939676

[43] Madiha Tabassum, Tomasz Kosiński, Alisa Frik, Nathan Malkin, Primal Wijesekera, Serge Egelman, and Heather Richter Lipford. 2019. Investigating Users' Preferences and Expectations for Always-Listening Voice Assistants. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3, 4 (2019), Article 153. https://doi.org/10.1145/3369807

[44] Nava Tintarev and Judith Masthoff. 2022. *Beyond Explaining Single Item Recommendations.* Springer US, New York, NY, 711–756.

[45] Suzanne Tolmeijer, Ujwal Gadiraju, Ramya Ghantasala, Akshit Gupta, and Abraham Bernstein. 2021. Second Chance for a First Impression? Trust Development in Intelligent System Interaction. In *Proceedings of the 29th ACM Conference on User Modeling, Adaptation and Personalization* (Utrecht, Netherlands) *(UMAP '21)*. Association for Computing Machinery, New York, NY, USA, 77–87. https://doi.org/10.1145/3450613.3456817

[46] Sarah Theres Völkel, Daniel Buschek, Malin Eiband, Benjamin R. Cowan, and Heinrich Hussmann. 2021. Eliciting and Analysing Users' Envisioned Dialogues with Perfect Voice Assistants. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 254, 15 pages.

[47] Nima Zargham, Leon Reicherts, Michael Bonfert, Sarah Theres Völkel, Johannes Schoening, Rainer Malaka, and Yvonne Rogers. 2022. Understanding Circumstances for Desirable Proactive Behaviour of Voice Assistants: The Proactivity Dilemma. In *Proceedings of the 4th Conference on Conversational User Interfaces* (Glasgow, United Kingdom) *(CUI '22)*. Association for Computing Machinery, New York, NY, USA, Article 3, 14 pages.